P1. Electrons: $-e$, $\mu_e = \left|\frac{e\tau_e}{m_e}\right|$, $n_e = n$

Holes: $+e$, $\mu_h = \left|\frac{e\tau_h}{m_h}\right|$, $n_h = p$.

In problem 3.3 (hw 03) we found that $\vec{j}_e = \tilde{\sigma}_e \vec{E}$

<u>For electrons</u> $\tilde{\sigma}_e = \frac{1}{\rho_{xx}^2 + \rho_{xy}^2}\begin{pmatrix} \rho_{xx} & \rho_{yx} \\ -\rho_{yx} & \rho_{xx} \end{pmatrix}$

where $\rho_{xx} = \frac{m_e}{n_e e^2 \tau_e}$, $\rho_{yx} = -\frac{B}{n_e e} = \frac{B}{n(-e)}$

$$\Rightarrow \rho_{xx} = \frac{m_e}{(-e)\tau_e}\frac{1}{(-e)n_e} = \frac{1}{\mu_e n e}$$

$$\rho_{xx}^2 + \rho_{xy}^2 = \frac{1}{\mu_e^2 n^2 e^2} + \frac{B^2}{n^2 e^2} = \frac{1 + B^2\mu_e^2}{\mu_e^2 n^2 e^2}$$

$$\Rightarrow \sigma_e = \frac{n^2 e^2 \mu_e^2}{1 + B^2\mu_e^2}\begin{pmatrix} \frac{1}{\mu_e n e} & \frac{-B}{ne} \\ \frac{B}{ne} & \frac{1}{\mu_e n e} \end{pmatrix} = \frac{n e \mu_e}{1 + B^2\mu_e^2}\begin{pmatrix} 1 & -\mu_e B \\ \mu_e B & 1 \end{pmatrix}$$

Therefore <u>for holes</u> we get $\sigma_h = \frac{p e \mu_h}{1 + B^2\mu_h^2}\begin{pmatrix} 1 & \mu_h B \\ -\mu_h B & 1 \end{pmatrix}$

(Since Lorentz force pushes holes opposite to electrons)

$\Rightarrow$ total current $\vec{j} = (\tilde{\sigma}_e + \tilde{\sigma}_h)\vec{E}$

$$\Rightarrow \vec{E} = \tilde{\sigma}^{-1}\vec{j} \Rightarrow \begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{pmatrix}\begin{pmatrix} j_x \\ 0 \end{pmatrix}$$

Since in Hall geometry $j_y = 0$

$$\Rightarrow \begin{cases} E_x = \rho_{xx} j_x \\ E_y = \rho_{yx} j_x \end{cases} \Rightarrow R_H = \frac{\rho_{yx}}{B}$$

$\Rightarrow$ we need to find inverse of $(\sigma_e + \sigma_h)$ matrix.

$$\rho^{-1} = \begin{pmatrix} \frac{ne\mu_e}{1+B^2\mu_e} & \frac{-ne\mu_e^2 B}{1+B^2\mu_e^2} \\ \frac{ne\mu_e^2 B}{1+B^2\mu_e^2} & \frac{ne\mu_e}{1+B^2\mu_e^2} \end{pmatrix} + \begin{pmatrix} \frac{pe\mu_h}{1+B^2\mu_h^2} & \frac{pe\mu_h^2 B}{1+B^2\mu_h^2} \\ -\frac{pe\mu_h^2 B}{1+B^2\mu_h^2} & \frac{pe\mu_h}{1+B^2\mu_h^2} \end{pmatrix} \equiv \begin{pmatrix} A & C \\ -C & A \end{pmatrix}$$

$$\Rightarrow \rho = \frac{1}{A^2+C^2} \begin{pmatrix} A & -C \\ C & A \end{pmatrix}$$

After some algebra

$$A^2+C^2 = \frac{1}{(1+B^2\mu_e^2)(1+B^2\mu_h^2)} \left[(ne\mu_e + pe\mu_h)^2 + (ne\mu_e B\mu_h - pe\mu_h B\mu_e)^2 \right]$$

so that $R_H = \frac{C}{A^2+C^2} \frac{1}{B}$

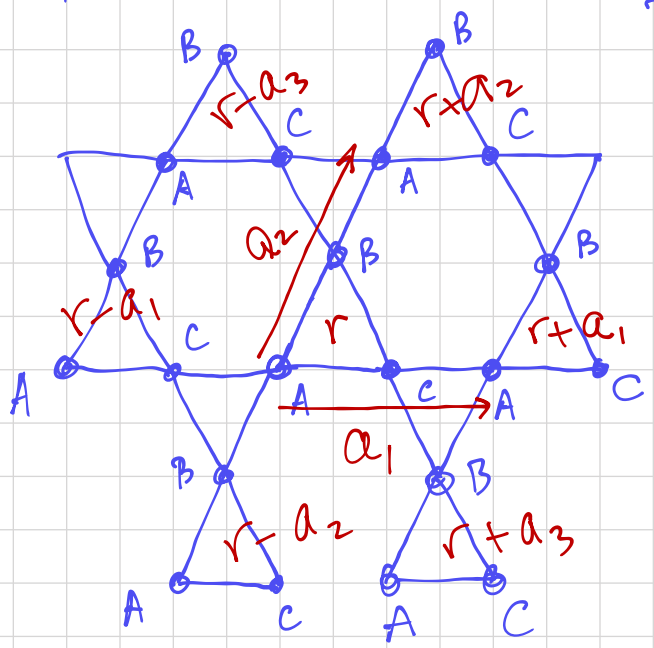
$$R_H = \frac{pe\mu_h^2(1+B^2\mu_e^2) - ne\mu_e^2(1+B^2\mu_h^2)}{(ne\mu_e + pe\mu_h)^2 + \mu_e^2\mu_h^2 B^2 (ne - pe)^2}$$

Low-field limit: $R_H = \frac{pe\mu_h^2 - ne\mu_e^2}{(ne\mu_e + pe\mu_h)^2}$

Easy check: for $\mu_h \to 0$ we get $R_H = \frac{-1}{ne}$

for $\mu_e \to 0$ $\quad R_H \to \frac{1}{pe}$.

P2. Electron bands of kagome lattice



$$\vec{a}_1 = a_0(1,0)$$
$$\vec{a}_2 = a_0\left(\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}\right)$$
$$\vec{a}_3 = \vec{a}_1 - \vec{a}_2 = a_0\left(\tfrac{1}{2}, -\tfrac{\sqrt{3}}{2}\right)$$
$$a_0 \equiv 2a$$

Start by writing out various hopping processes in real space.
For example, $(-t)\,|\vec{r},A\rangle\langle\vec{r},B|$ describes electron from site B in the unit cell $\vec{r}$ hopping to site A in the same unit cell.

Focusing on all processes that end up at $|\vec{r},A\rangle$ we can write

$$(-t)\,|\vec{r},A\rangle\Big(\langle\vec{r},B| + \langle\vec{r},C| + \langle\vec{r}-\vec{a}_2,B| + \langle\vec{r}-\vec{a}_1,C|\Big) \quad (1)$$

In second quantization, this reads

$$\boxed{-t\,\hat{a}^{+}_{r}\left(\hat{b}_r + \hat{c}_r + \hat{b}_{r-a_2} + \hat{c}_{r-a_1}\right.}$$

Similarly, we have

$$(-t)\,|\vec{r},B\rangle\Big(\langle\vec{r},A| + \langle\vec{r},C| + \langle\vec{r}-\vec{a}_3,C| + \langle\vec{r}+\vec{a}_2,A|\Big) \quad (2)$$

and

$$(-t)\,|\vec{r},C\rangle\Big(\langle\vec{r},A| + \langle\vec{r},B| + \langle\vec{r}+\vec{a}_3,B| + \langle\vec{r}+\vec{a}_1,A|\Big) \quad (3)$$

The full Hamiltonian should be hermitian and can be written as

$$\hat{H} = \{(1) + (2) + (3)\} + h.c.$$

↳ hermitian conjugate of { } part.

Next, Fourier transform and write in matrix form

$$\hat{H} = (-t)\left(|\vec{k},A\rangle, |\vec{k},B\rangle, |\vec{k},C\rangle\right) \underbrace{\begin{pmatrix} 0 & 1+e^{ik_2} & 1+e^{ik_1} \\ 1+e^{-ik_2} & 0 & 1+e^{ik_3} \\ 1+e^{-ik_1} & 1+e^{-ik_3} & 0 \end{pmatrix}}_{M} \begin{pmatrix} \langle\vec{k},A| \\ \langle\vec{k},B| \\ \langle\vec{k},C| \end{pmatrix}$$

here $k_j \equiv \vec{k}\cdot\vec{a}_j$ $(j=1,2,3)$

To find eigenvalues we need to solve

$$\det\left[-tM - E\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right] = 0$$

Mathematica gives

$$(E-2t)\left\{E^2 + 2tE - t^2\left[2 + 2\cos k_1 + 2\cos k_2 + 2\cos k_3\right]\right\} = 0$$

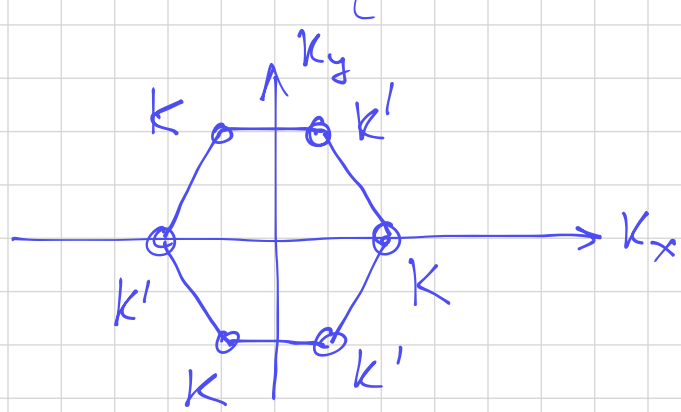
Thus there is a <u>flat</u> band $E_3(\vec{k}) = 2t$.

and two dispersing bands

$$E_{1,2}(\vec{k}) = -t \pm t\sqrt{1+2A_k}$$

$$A_k = 1 + \cos(k_x a_0) + 2\cos\left(\frac{k_x a_0}{2}\right)\cos\left(\frac{\sqrt{3}\,k_y a_0}{2}\right).$$

So that $E_{1,2}(\vec{k}) = -t \pm t\left\{3 + 2\cos(k_x a_0) + 4\cos\left(\frac{k_x a_0}{2}\right)\cos\left(\frac{\sqrt{3}k_y a_0}{2}\right)\right\}^{1/2}$



$$K = \left(\frac{4\pi}{3a_0}, 0\right)$$

Observe that at $\vec{k} = K = \left(\frac{4\pi}{3a_0}, 0\right)$

$$\sqrt{1+2A_k} = \sqrt{3 + 2\cos\left(\frac{4\pi}{3}\right) + 4\cos\left(\frac{2\pi}{3}\right)} = 0,$$ and similarly at K'.

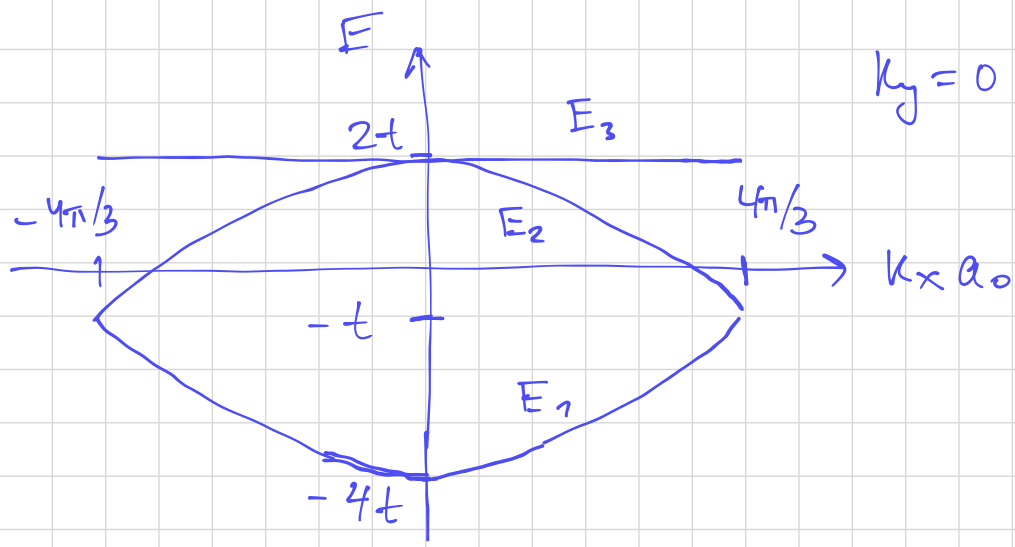
Therefore $E_1 = -t - t\sqrt{1+2A_k}$

and $E_2 = -t + t\sqrt{1+2A_k}$

touch each other at K and K' points.

At Γ point $(\vec{k}=(0,0))$ $\sqrt{1+2A_k} = \sqrt{9} = 3$

$$\Rightarrow \begin{cases} E_1 = -t - 3t = -4t \\ E_2 = -t + 3t = 2t \end{cases} \leftarrow E_2 \text{ touches } E_3 \text{ at } \Gamma.$$



To find effective mass, we can expand

$$\sqrt{1+2A_k} \xrightarrow[\substack{k_x \to 0 \\ k_y \to 0}]{} 3 - \frac{1}{4}\left(k_x^2 + k_y^2\right) a_0^2$$

Therefore
$$\begin{cases} E_1(k) \approx -4t + \frac{1}{4} t a_0^2 \vec{k}^2 = -4t + \frac{\hbar^2 \vec{k}^2}{2m_1} \\ E_2(k) \approx 2t - \frac{1}{4} t a_0^2 \vec{k}^2 = 2t + \frac{\hbar^2 \vec{k}^2}{2m_2} \\ E_3(k) = 2t = 2t + \frac{\hbar^2 \vec{k}^2}{2m_3} \end{cases}$$

Therefore <u>electron</u> effective mass is (near $\vec{k}=0,0$)

$$\begin{cases} m_1 = \frac{2\hbar^2}{t a_0^2} & \text{on band } E_1 \\ m_2 = -\frac{2\hbar^2}{t a_0^2} & \text{on band } E_2 \\ m_3 = \infty & \text{on band } E_3 \end{cases}$$